brown open silicon

designing open chips. together.

agenda

icebreaker

spring schedule

IEEE code-a-chip, chipathon

projects, next steps

School of Engineering
BROWN UNIVERSITY

icebreaker

1 “small” super power

for example: being able to know the right orientation to turn a shower handle in a hotel the first time

name + year + concentration

School *of* Engineering
BROWN UNIVERSITY

spring schedule

Starting February 7th)	Meeting Goals
Week 1 (Feb 21nd)	Open hours for questions + support with setup
Week 2 (Feb 28th)	Project Selection Project Ideation/ Team Forming
Week 3 (Mar 7th)	Anhang tool demonstrations (Both Custom + Digital) Manufacturing Process (just for general knowledge) Transistor Sizing NMOS & PMOS basic "architecture" Stick Diagrams + Layout Tips (Anhang) + How to read DRC and LVS Reverse engineering silicon with Anhang
Week 4 (Mar 14th)	Industry Chat (**APPLE**)
Week 5 (Apr 4th)	How to debug session (Custom -> Anhang/Ruichen) (Digital -> Xinting)
Week 6 (Apr 11th)	Design Checkup/Support
Week 7 (Apr 18th)	Open hours for questions + support
Week 8 (Apr 25th)	Industry Chat (TBD)
Week 9 (May 2nd)	Careers in Chip Design + Possible Research w/ Prof. Saligane's Lab Support + Leadership Roles for future (anyone interested in helping run it)
Week 10 (May 9th)	Wrap up meeting, feedback, food!

School *of* Engineering
BROWN UNIVERSITY

IEEE code-a-chip

design a chip within jupyter notebook that can be run by anyone!

Selects up to 10 winners and travel grants are up to $2,500 (undergraduates), and up to $5,000 (graduate students)

School of Engineering
BROWN UNIVERSITY

chipathon

summer long event aimed at developing a chip and taping out using fully open-source tools

School *of* Engineering
BROWN UNIVERSITY

available projects (tentative)

self-defined projects

look for a target problem/application

form groups of interest

design idea & high-level architecture

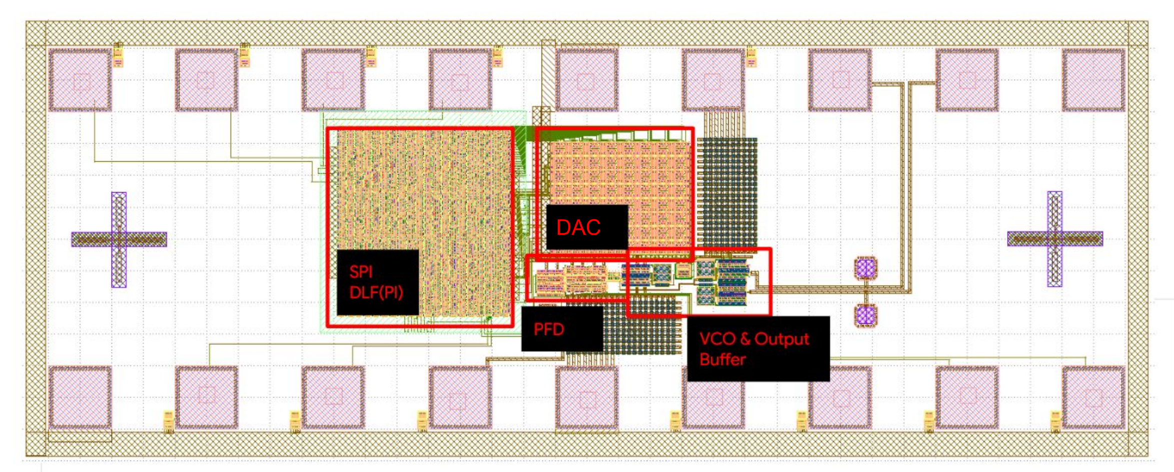
digital + custom

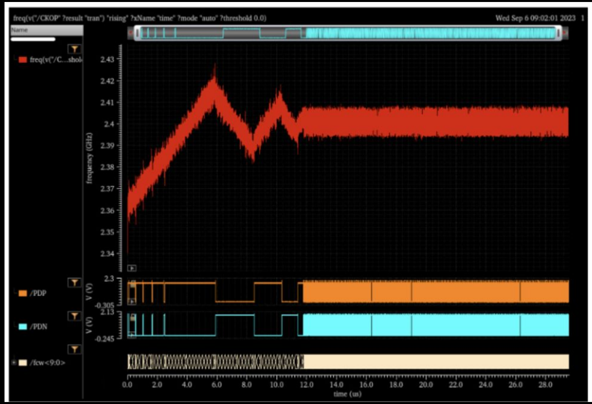
digital PLL with custom-designed VCO

School *of* Engineering
BROWN UNIVERSITY

template project

digital + custom design: PLL(Phased-locked Loop)



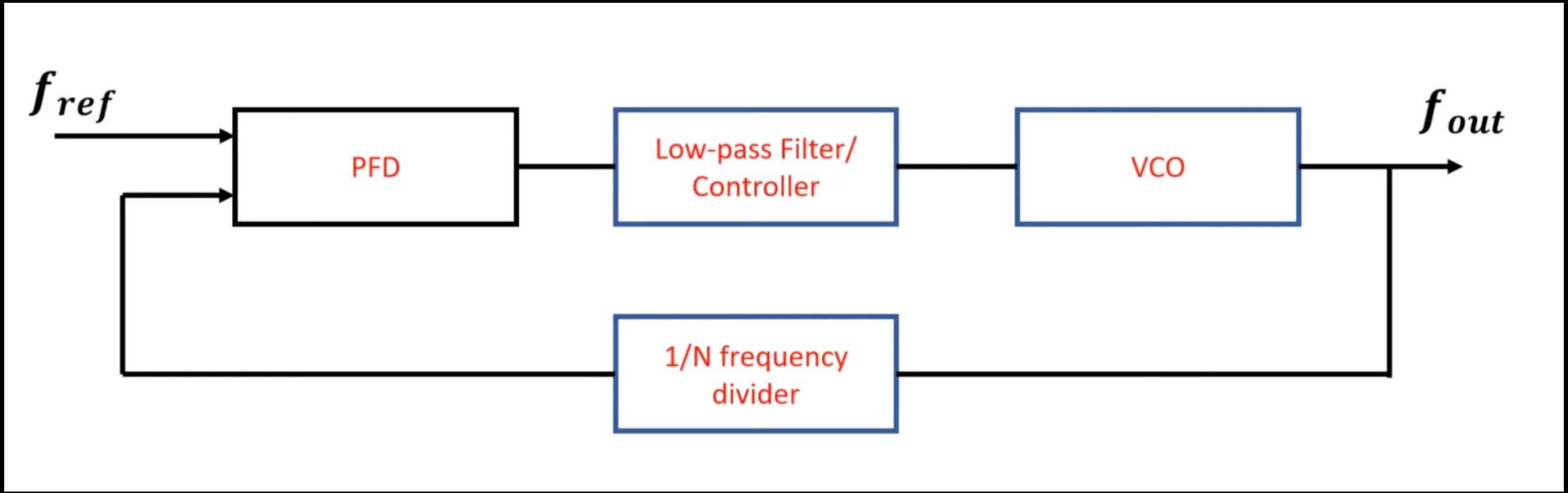


Input : 240MHz Signal
Output: 2.4GHz Signal

School of Engineering
BROWN UNIVERSITY

template project

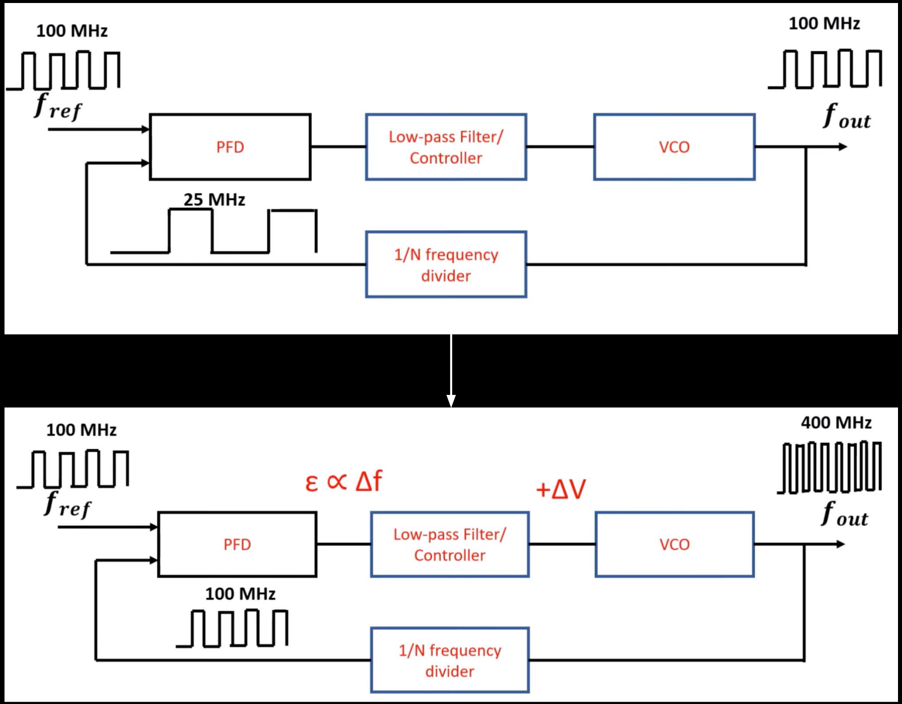
digital + custom design: PLL(Phased-locked Loop)



School of Engineering
BROWN UNIVERSITY

template project

digital + custom design: PLL(Phased-locked Loop)



template project

digital + custom design: PLL(Phased-locked Loop)

Great video explaining the full workings of a PLL and can help you get started!

We will post digital design flow tutorials soon!

self-defined projects

School *of* Engineering
BROWN UNIVERSITY

next steps

forming teams
begin high-level architecture design
start building! we're here to help

administrative support needed

design poster for club + speaker events
create a media channel for resources/tool support

School *of* Engineering
BROWN UNIVERSITY

thank you for coming!

School *of* Engineering
BROWN UNIVERSITY